



Congratulations on your selection as a semi-finalist for our Applied Biomechanics Quantitative Analyst role. As a means to learn more about each candidate, we request that you complete this questionnaire to the best of your ability within 5 days of your receipt, unless otherwise instructed in our email.

Prompt: There are two datasets attached. One contains pitcher biomechanics and pitch velocities for a sample of pitches thrown by lower level minor league pitchers in 2024 and 2025 (`velo_and_mechanics.csv`). The other contains only the biomechanics data (no measured pitch velocities) for a distinct set of pitchers in the Florida Complex League (FCL) in 2025. Use this data to answer the question: out of the provided pitchers in the 2025 FCL dataset, which pitcher do you believe will throw the hardest in five years? You may construct any model(s) or metric(s) to support your conclusion.

Deliverables: Your response should include:

- a clear description of your methodology, including all data processing and/or modeling steps,
- any quantitative and/or qualitative tests that you used to validate your approach, and
- all of the code that you used in preparing your solution.

You may use any programming language and presentation format (e.g. slides, writeup, notebook) to prepare your solution.

Data: We have attached three CSVs, which are described below.

- `velo_and_mechanics.csv`: the pitch velocity and pitcher biomechanics for a sample of 2024 fastballs in the FCL (rookie ball), and 2025 fastballs in the FSL (single-A).
- `fcl_mechanics_2025.csv`: pitcher biomechanics for a sample of 2025 fastballs in the FCL.
- `metric_descriptions.csv`: descriptions of the columns provided in the above CSVs.